

WEEK 1 · DAY 4

Cloud Economics Overview

BaseStack Academy · AWS Cloud Accelerator

Cloud Foundations

Today's Agenda

Core AWS Services Overview

1 Overview — Why This Matters

FOUNDATION

2 Key Concepts — Definitions & Vocabulary

CORE KNOWLEDGE

3 Deep Dive — Key Concepts Extended

DEEPER LEARNING

4 How It Works — Step by Step

MECHANICS

5 Technical Deep Dive

ADVANCED DETAIL

6 Architecture & Visual Model

VISUAL

7 Real-World Nigerian Scenario

CONTEXT

8 Common Mistakes & Exam Traps

EXAM PREP

Overview — Why This Matters

1

Traditional IT = Upfront Capital (CapEx)

Buying servers for peak demand — even if that peak lasted just 3 days a year. Massive upfront investment, mostly idle.

2

Cloud Converts CapEx to OpEx

Pay only for what you use — per second, per request. No idle hardware. Cash flow improves dramatically.

3

The African Business Advantage

A Lagos startup can launch production-ready on AWS with ₦50,000 in credits. Scale spend only as customers grow.

WHY NOW?

☁ Cloud = #1 in-demand tech skill globally

🎯 SAA-C03 targets ₦5M+ salary roles

🚀 Built by Paystack, Flutterwave, Piggyvest

💡 No CapEx → FinTechs launch in days, not months

💡 Beginner Note

Do not worry if this feels new — every expert was once a complete beginner. Take notes and ask questions freely.



Key Concepts — Know These Definitions

📖 Study tip: Say each definition aloud — spoken repetition builds memory 3× faster than reading alone.

Pay-As-You-Go

No upfront cost, no termination fees, pay per second/hour/request of actual usage

Reserved Instances

1 or 3-year commitment in exchange for up to 72% discount vs On-Demand pricing

Spot Instances

Use spare AWS capacity at up to 90% discount — but can be interrupted with 2 minutes notice

Savings Plans

Flexible discount model that applies across EC2, Lambda, and Fargate when you commit to a spend level

AWS Free Tier

Three types — 12-month free, always free, and short-term trials for new services



Key Concepts — Quick Review

\$0

Pay-As-You-Go

No upfront cost. No termination fees.
Billed per second, hour, or request.

72%

Reserved Instances

1 or 3-year instance commitment for up to
72% discount vs On-Demand.

90%

Spot Instances

Use spare AWS capacity at up to 90% off —
can be interrupted with 2-min notice.

66%

Savings Plans

Commit to a spend level (not instance
type) for discounts across EC2, Lambda,
Fargate.

750hrs

AWS Free Tier

EC2 t2.micro for 12 months + 1M Lambda
requests + 5GB S3 — always free.

⚙️ How It Works — Step by Step

1

How AWS Bills You

Each service uses its own billing metric. EC2 charges per second (min. 60s) based on instance type and Region. S3 charges per GB stored per month plus per-request fees. Lambda bills per invocation and per millisecond of execution time.

KEY INSIGHT: You only pay for active resources — stop an EC2 instance and billing stops.

2

The AWS Free Tier — Your Safety Net

EC2 t2.micro or t3.micro: 750 hours/month free (12 months) · S3: 5GB storage + 20,000 GET + 2,000 PUT requests/month, always free · Lambda: 1 million requests + 400,000 GB-seconds of compute/month, always free

KEY INSIGHT: You can complete the entire SAA-C03 lab curriculum without spending a naira.

3

Monitor & Control Your Costs

AWS Cost Explorer: historical spend analysis with 13-month forecasting. AWS Budgets: set alerts for actual or forecasted spend before you go over. AWS Pricing Calculator: estimate costs before you build.

KEY INSIGHT: Set a ₦0 budget alert in AWS Budgets on Day 1 of your account.



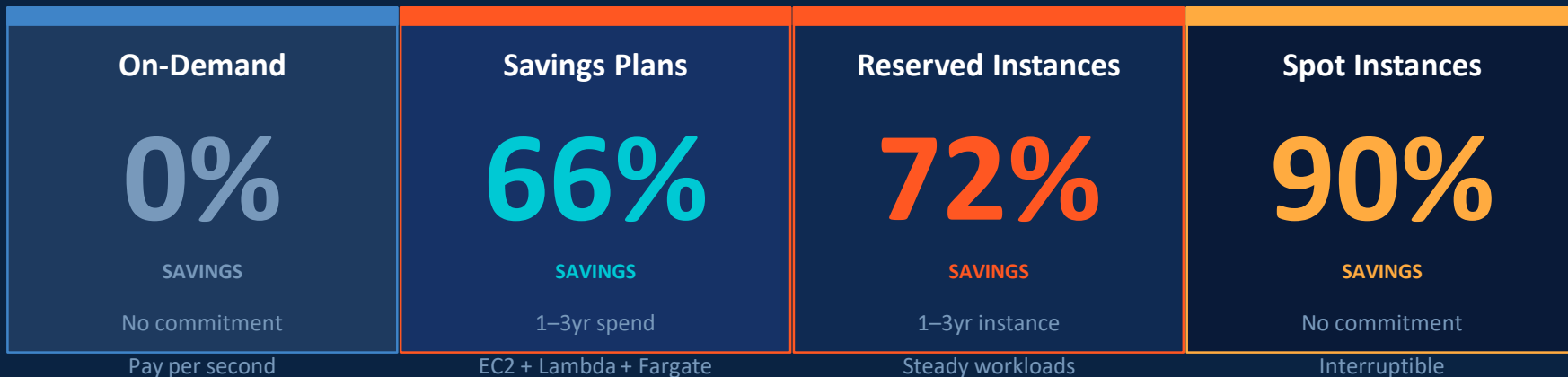
AWS Pricing Models — Side-by-Side Comparison

Pricing Model	Commitment	Max Discount	Best For	⚠️ Exam Watch
On-Demand	None	0%	Dev/test, unpredictable workloads	Default — baseline pricing
Savings Plans	1–3 yr (spend)	66%	Flexible compute: EC2 + Lambda + Fargate	Commit to \$ spend, not instance type
Reserved Instances	1–3 yr (instance)	72%	Steady-state, predictable workloads	Instance type locked — less flexible
Spot Instances	None	90%	Fault-tolerant & interruptible jobs	2-min termination notice — NEVER for DBs
AWS Free Tier	N/A	100%*	Learning & new service trials	3 types: 12-month, always-free, trial

❓ Architecture — AWS Pricing Model Spectrum

← More Flexible | Lower Savings

Higher Savings | More Commitment →



AWS FREE TIER

750 hrs EC2 t2.micro · 1M Lambda requests · 5GB S3 — All free

⚠️ EXAM TRAP: Spot Instances can be terminated with just 2-minute notice — NEVER use for primary databases or stateful apps.



Real-World Nigerian Scenario — Paystack on AWS

✗ BEFORE: Physical Servers

\$73,000+

YEAR ONE INFRASTRUCTURE COST

\$50,000

Server purchase (production-grade)

\$15,000/yr

Data centre colocation fees

\$8,000/yr

Dedicated sysadmin salary

✓ AFTER: On AWS

~\$300/mo

TO START — SCALES WITH GROWTH

✓ Pay per transaction — minimal cost at 3am

✓ Auto-scales for Black Friday payment surges

✓ AWS manages infra — team focuses on product

💬 Discussion Question

Think about a Nigerian business you know. How would this AWS service change how they operate? What problem would it solve?

⚠️ Common Mistakes & Exam Traps

- These are the most common exam errors — every one costs marks. Memorise them.



Free Tier has THREE types: 12-month free (e.g., EC2 t2.micro), always-free (e.g., Lambda 1M requests), and short-term trials (e.g., Amazon Redshift 2-month trial). Knowing which type applies is an exam question.



Spot Instances can be interrupted by AWS with only a 2-minute warning — NEVER use them for stateful applications, primary databases, or jobs that cannot tolerate interruption.



Data transfer WITHIN the same Region between AZs incurs charges (small). Data transfer between Regions incurs larger charges. Always account for data transfer in architecture cost estimates.



Question

A Solutions Architect at a Lagos fintech company is designing the AWS infrastructure for a new payment platform. They need to implement best practices related to Core AWS Services Overview. Which of the following statements is CORRECT?

A

No upfront cost, no termination fees, pay per second/hour/request of actual usage

✓ CORRECT

B

Free Tier has THREE types: 12-month free (e.g., EC2 t2.micro), always-free (e.g., Lambda 1M...

C

Spot Instances can be interrupted by AWS with only a 2-minute warning — NEVER use them for...

D

Data transfer WITHIN the same Region between AZs incurs charges (small). Data transfer between...

💡 Key Takeaways — What You Learned Today

Core AWS Services Overview

✓ Concepts Covered

✓ Pay-As-You-Go

✓ Reserved Instances

✓ Spot Instances

✓ Savings Plans

✓ AWS Free Tier

📁 Before Next Class

1. Review the concepts above without notes

2. Complete the hands-on console lab

3. Post your progress on LinkedIn / X

4. Review the exam traps from the previous slide

5. Check the Further Resources list



Further Resources & Reading

Go deeper with these official AWS resources — read at least ONE before next class:

AWS Cloud Pricing Overview

aws.amazon.com/pricing

The master reference for pricing across all AWS services. Bookmark this.

AWS Free Tier Dashboard

aws.amazon.com/free

Track your 12-month free, always-free, and trial limits. Essential reading.

AWS Savings Plans

aws.amazon.com/savingsplans

Flexible discount model covering EC2, Lambda, and Fargate in one commitment.

AWS Cost Explorer

aws.amazon.com/aws-cost-management/aws-cost-explorer

Visualise and forecast your AWS spend over 13 months. Free to use.

AWS Budgets

aws.amazon.com/aws-cost-management/aws-budgets

Set spend alerts before you exceed your budget. Set this up on Day 1.

AWS Pricing Calculator

calculator.aws

Estimate your monthly AWS bill before committing. Build and share estimates.



Homework: Read at least ONE link above before the next class. Post what you learned on the WhatsApp group.



BASE STACK

A C A D E M Y

Week 1 · Day 4 Complete

Cloud Economics Overview

Cloud Foundations

Next: Hands-on Console Lab →